AutoDESN with a normalized regularized horseshoe

Implementation, independent validation and frozen C1 benchmark | October 22, 2026

Repository: AntonioAPDL/exdqlm. Base revision f397cdcd6df66e13f012bfbd3d299417708dfc24. Draft pull request #3. This report is generated only after the three-model audit has passed.

Pooled predictive performance

model	RMSE	MAE	CRPS	NLL
AutoDESN ridge	0.6476	0.4632	0.3501	0.9818
AutoDESN RHS	0.6487	0.4644	0.3508	0.9835
AutoGP	0.6177	0.4323	0.3404	0.9777

model	Coverage50	Coverage90	Width50	Width90
AutoDESN ridge	0.6536	0.9033	0.9286	2.2621
AutoDESN RHS	0.6524	0.9032	0.9309	2.2684
AutoGP	0.7129	0.9361	1.0339	2.5259

Scores describe one previously selected synthetic series. The 14,130 overlapping origin-horizon pairs per model are dependent, reuse 500 outcomes, and are not independent simulation replications. No general statistical superiority claim is supported.

Frozen comparison

500 conditioning/washout observations, followed by exactly 2,000 training responses (501-2500), then 500 test responses. Fit once at 2500; issue 30-step forecasts at all 471 origins from 2500 through 2970. All parameters, architecture weights, scaling and reservoirs stay fixed. Newly observed responses update states between origins; simulated responses feed back within each issued DESN forecast.

The exact original C1 data and 456 candidate definitions were recovered, with SHA256 checks. The original initial-state washout rule admits 428 candidates. Ridge and RHS use the identical bank and original feature coordinates. No test outcomes select priors or the frozen computational MH proposal.

Statistical target and exact updates

For architecture m, X_m=[1, H_m/sqrt(d_m)] and y | beta_m, sigma2 ~ N(X_m beta_m, sigma2 I). The intercept is N(0,10). Observation variance is IG(3,1), independently of coefficients. Ridge slopes retain their original learned common precision. RHS slopes use variance v_j=(1/c2+1/(tau0_m*u*lambda_j)^2)^(-1), with u and lambda_j independent standard half-Cauchy, tau0=1, and c2 ~ IG(2,8).

Every Gaussian coefficient density includes (2*pi*v_j)^(-1/2). Integrating coefficients gives log p(y | m, scales, sigma2) = -0.5[n log(2*pi)+log|C_m|+y' C_m^(-1)y], where C_m=sigma2 I+X_m D_m X_m'. Stable prior-whitened Cholesky systems evaluate this integral in the original feature basis; RHS is not rotation invariant.

An auxiliary vector of d_max proper half-Cauchy local scales couples models computationally. Model m uses its first d_m entries; inactive entries integrate to one and are refreshed from their priors. The architecture update collapses coefficients and uses an exact Hastings ratio containing prior odds, integrated likelihood odds, and reverse/forward proposal odds. Coefficients are immediately redrawn in the selected dimension, followed by variance and scale updates.

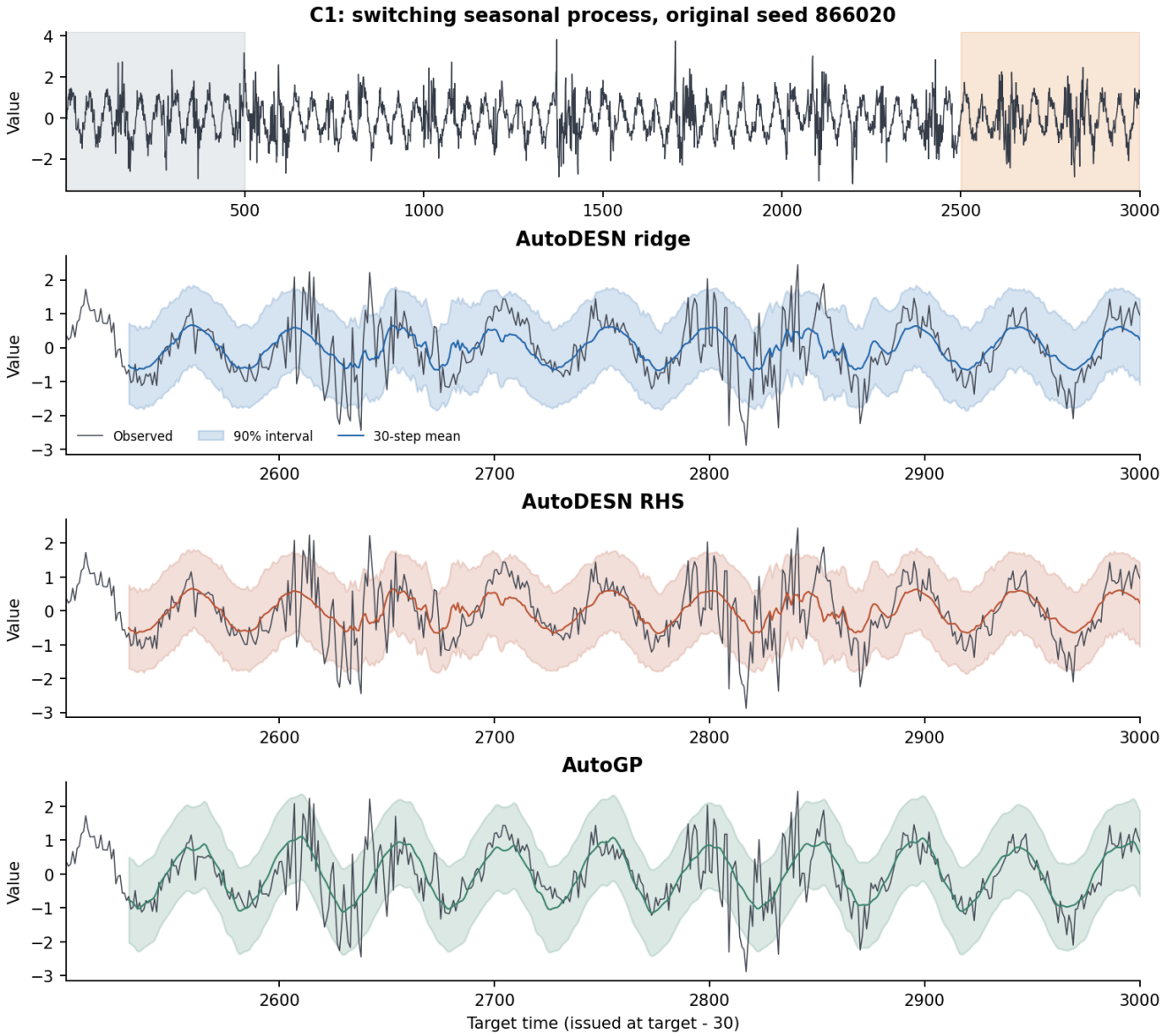
Given scales, beta has precision X'X/sigma2+D^(-1), mean equal to its inverse times X'y/sigma2. Noise variance is IG(3+n/2,1+||y-X beta||^2/2). The exact local rejection kernel uses a two-component inverse-gamma envelope with acceptance at least 1/sqrt(2). Global and slab scales use unbounded log-coordinate slice updates, including both Jacobians.

Why rhs_ns is a different target

The existing rhs variational block uses the normalized coefficient variance. The rhs_ns block multiplies a horseshoe coefficient density by a Gaussian pseudo-observation. N(beta;0,a) N(0;beta,c2) equals N(beta;0,a*c2/(a+c2)) N(0;0,a+c2). That extra scale-dependent factor changes scale conditionals and generally introduces a dimension-dependent integrated normalizer. Its conjugate updates cannot be substituted into this normalized RHS implementation.

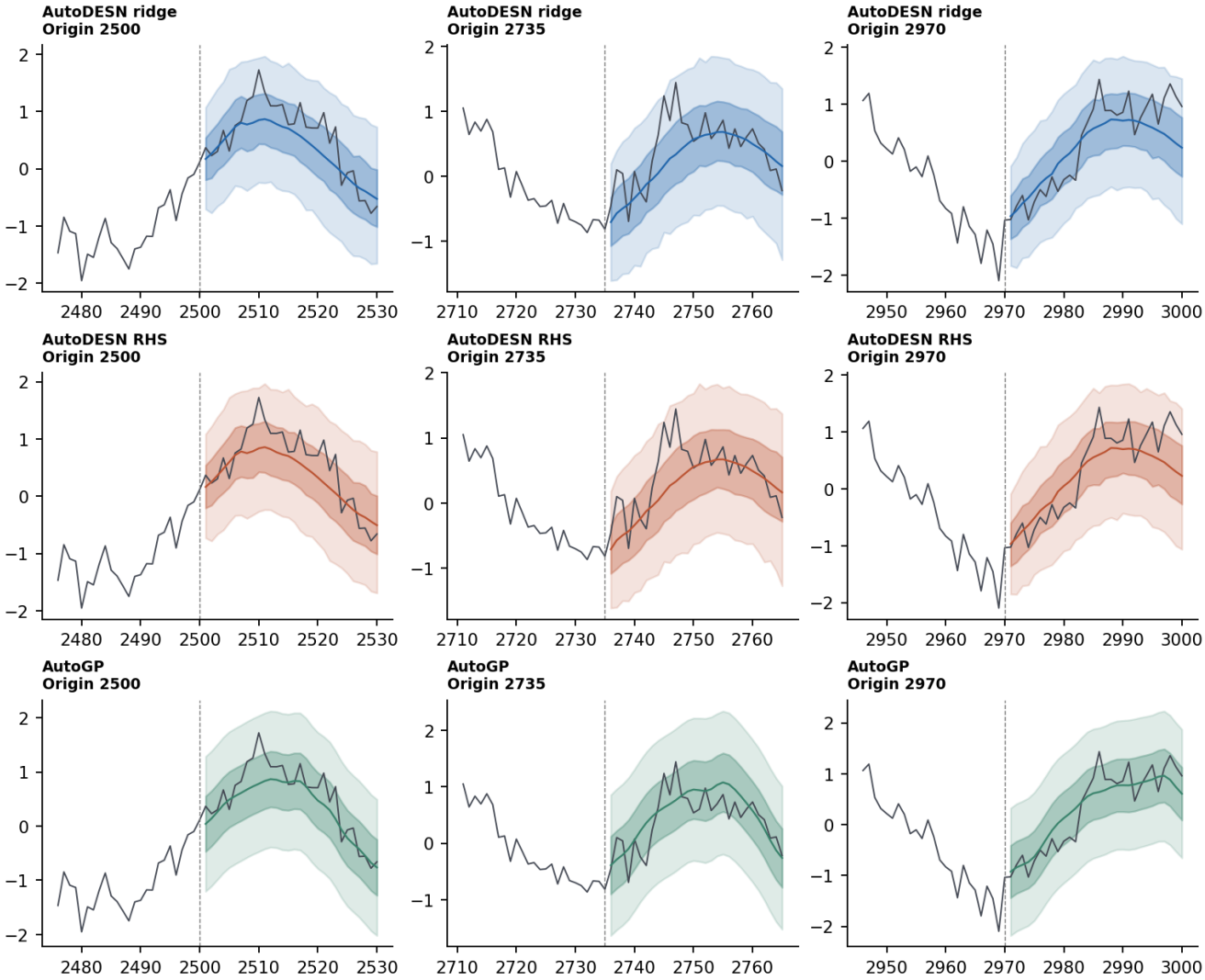
Full derivations and API controls are in docs/autodesn/RHS.md. Primary reference: Piironen and Vehtari (2017), Electronic Journal of Statistics, DOI 10.1214/17-EJS1337SI, arXiv:1707.01694. The Gaussian likelihood is conditional on causal autoregressive feature rows; its coefficient integral does not assert that the full recursively generated series is jointly Gaussian.

Causal 30-step rolling forecasts



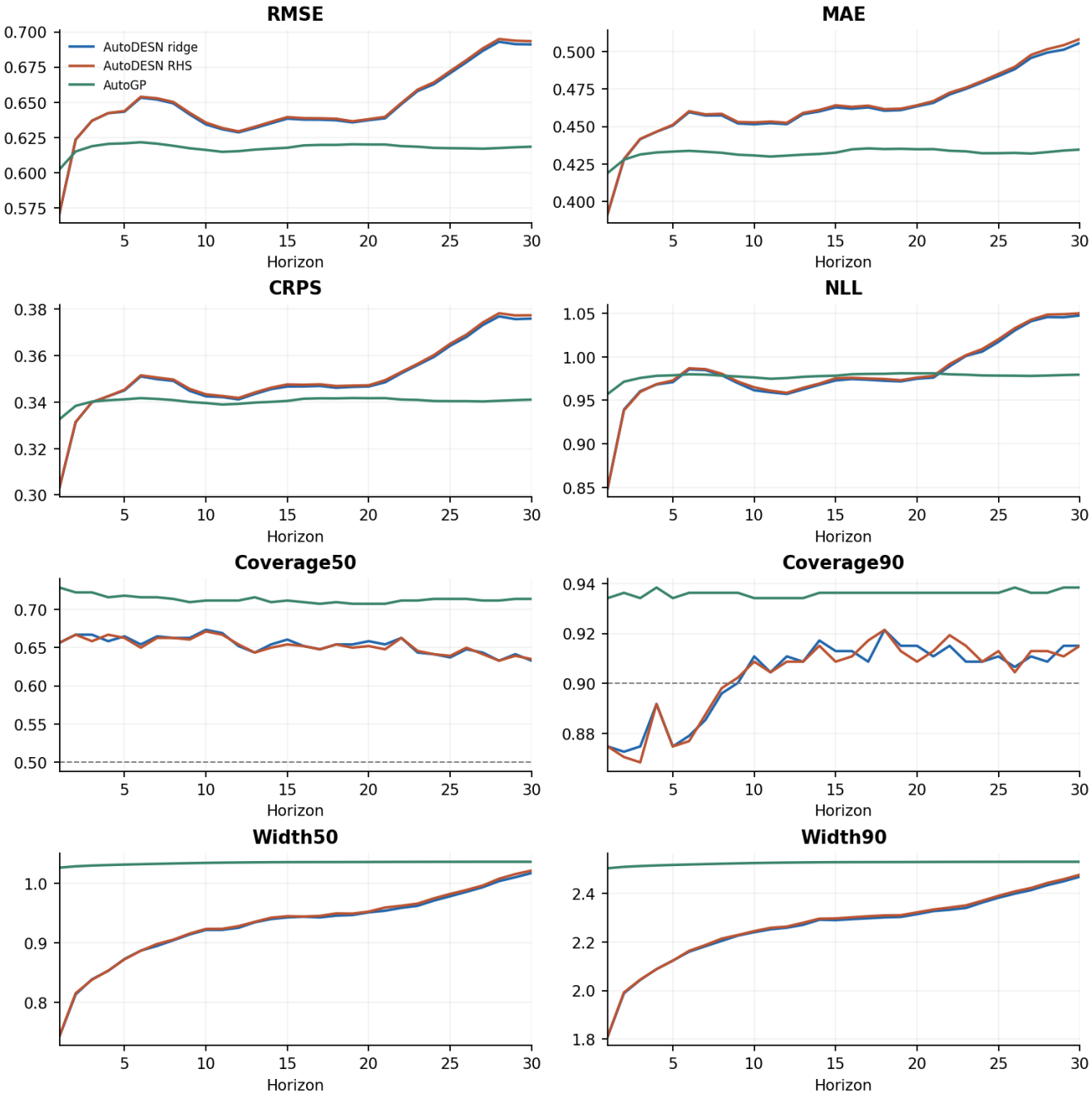
Each plotted prediction was issued 30 observations before its target. Intervals use individual predictive paths, preserving a joint architecture/coefficient/variance draw through all horizons. Responses unavailable at issue time never enter the forecast engine.

Three issued forecast windows



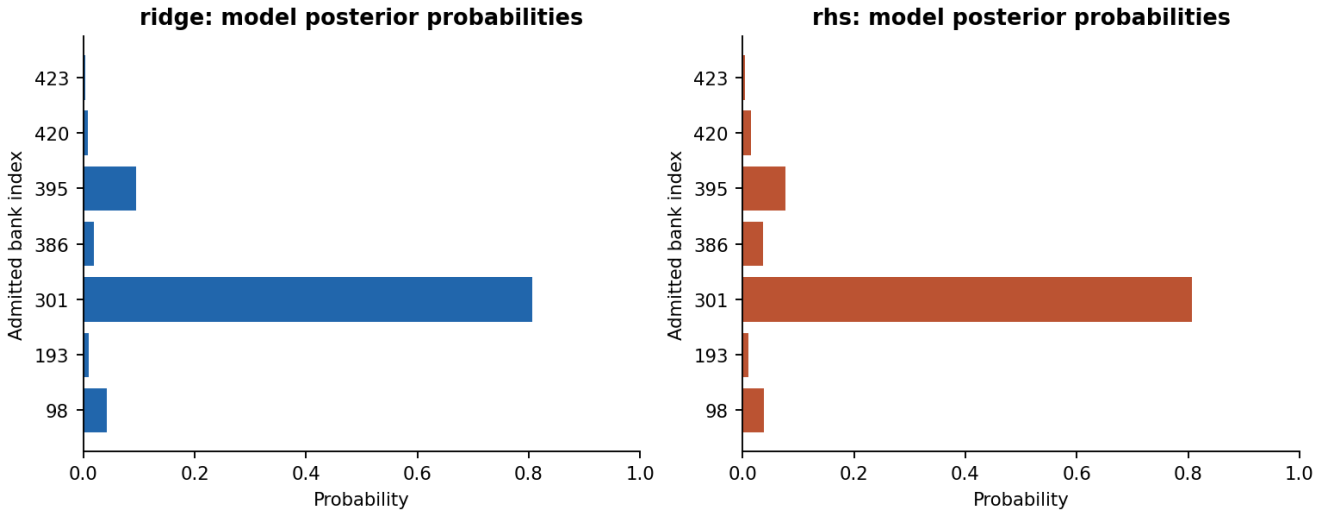
Grey curves show observations. The dashed line marks issue time; darker and lighter bands are 50% and 90% intervals. Forecasts from different origins are never pooled into a single interval.

All metrics by forecast horizon



Lower error, CRPS and NLL are better. Coverage should be assessed together with interval width. DESN CRPS is estimated from simulated predictive paths; AutoGP CRPS uses its analytic Gaussian mixture. NLL uses Gaussian-mixture log densities. Monte Carlo sensitivity is reported separately.

Posterior architecture averaging



RIDGE

model	candidate	D	posterior_probability	chain_range
301	broad_236	3	0.8061	0.0171
395	broad_343	4	0.0942	0.0063
98	broad_026	1	0.0425	0.0074
386	broad_332	4	0.0193	0.0004
193	broad_122	2	0.0098	0.0019

RHS

model	candidate	D	posterior_probability	chain_range
301	broad_236	3	0.8075	0.0277
395	broad_343	4	0.0765	0.0042
98	broad_026	1	0.0375	0.0045
386	broad_332	4	0.0367	0.0074
420	broad_375	4	0.0150	0.0095

Candidate IDs index the same admitted bank. Probabilities average over the posterior rather than selecting one screened architecture. An unvisited low-probability model does not have proven zero mass.

MCMC convergence and exploration

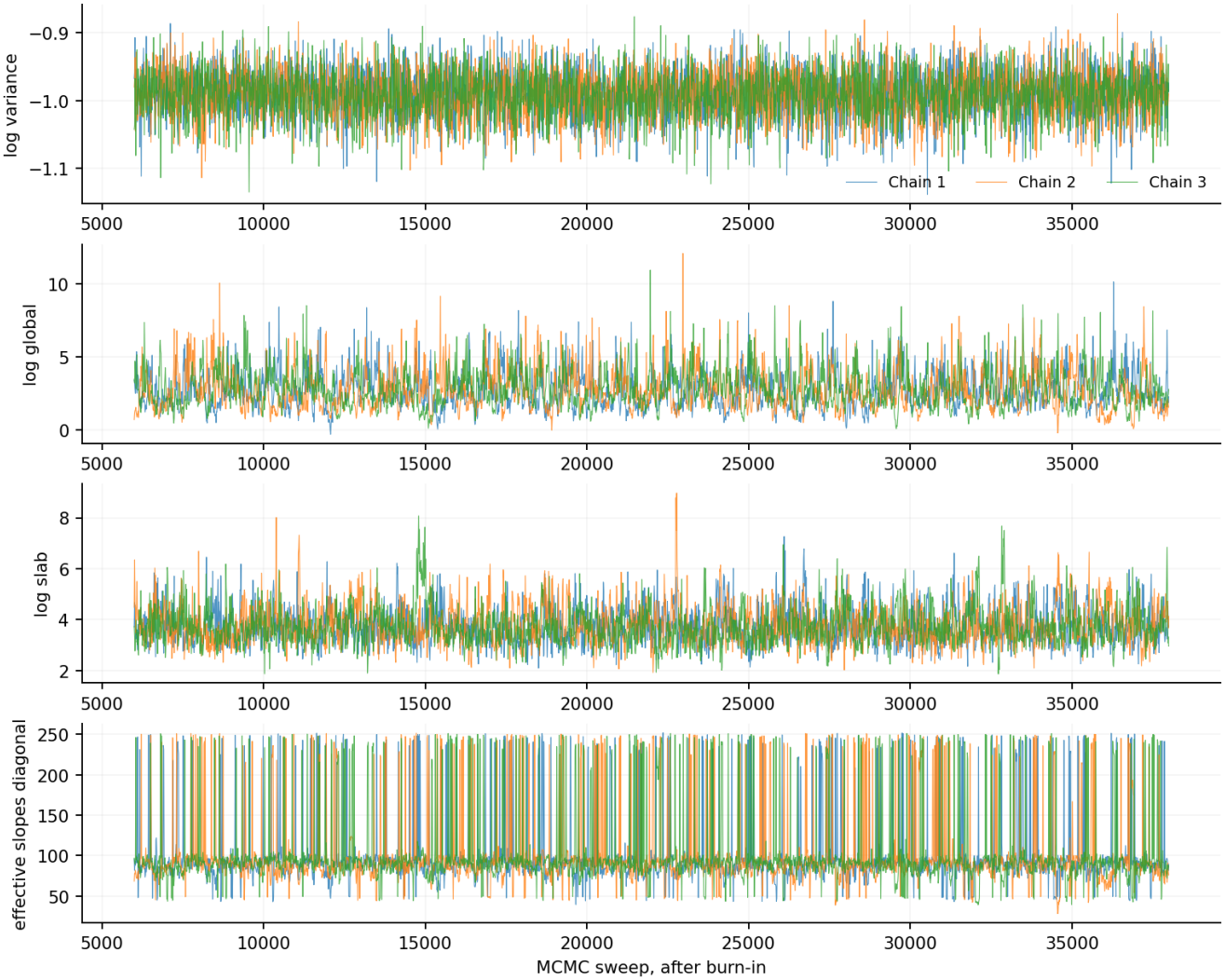
model	parameter	rhat	ess_spectral
ridge	log_variance	1.0002	11006.1722
ridge	log_ridge_precision	1.0008	521.9809
rhs	log_variance	1.0001	32216.9098
rhs	log_global	1.0021	1148.6099
rhs	log_slab	1.0014	1998.8201
rhs	mean_log_local	1.0001	34623.0808
rhs	intercept	1.0008	20770.2030
rhs	fitted_rms	1.0001	65823.1833
rhs	effective_slopes_diagonal	1.0018	6117.4145

method	chain	retained	unique_models	retained_switches
ridge	1	6000	16	1440
ridge	2	6000	16	1437
ridge	3	6000	18	1564
rhs	1	32000	15	2518
rhs	2	32000	17	2442
rhs	3	32000	14	2611

An independent FFT-based check also evaluates bulk/tail ESS and indicators for models with at least 1% posterior mass. Maximum R-hat: 1.0026; minimum bulk ESS: 558.5; minimum tail ESS: 656.9.

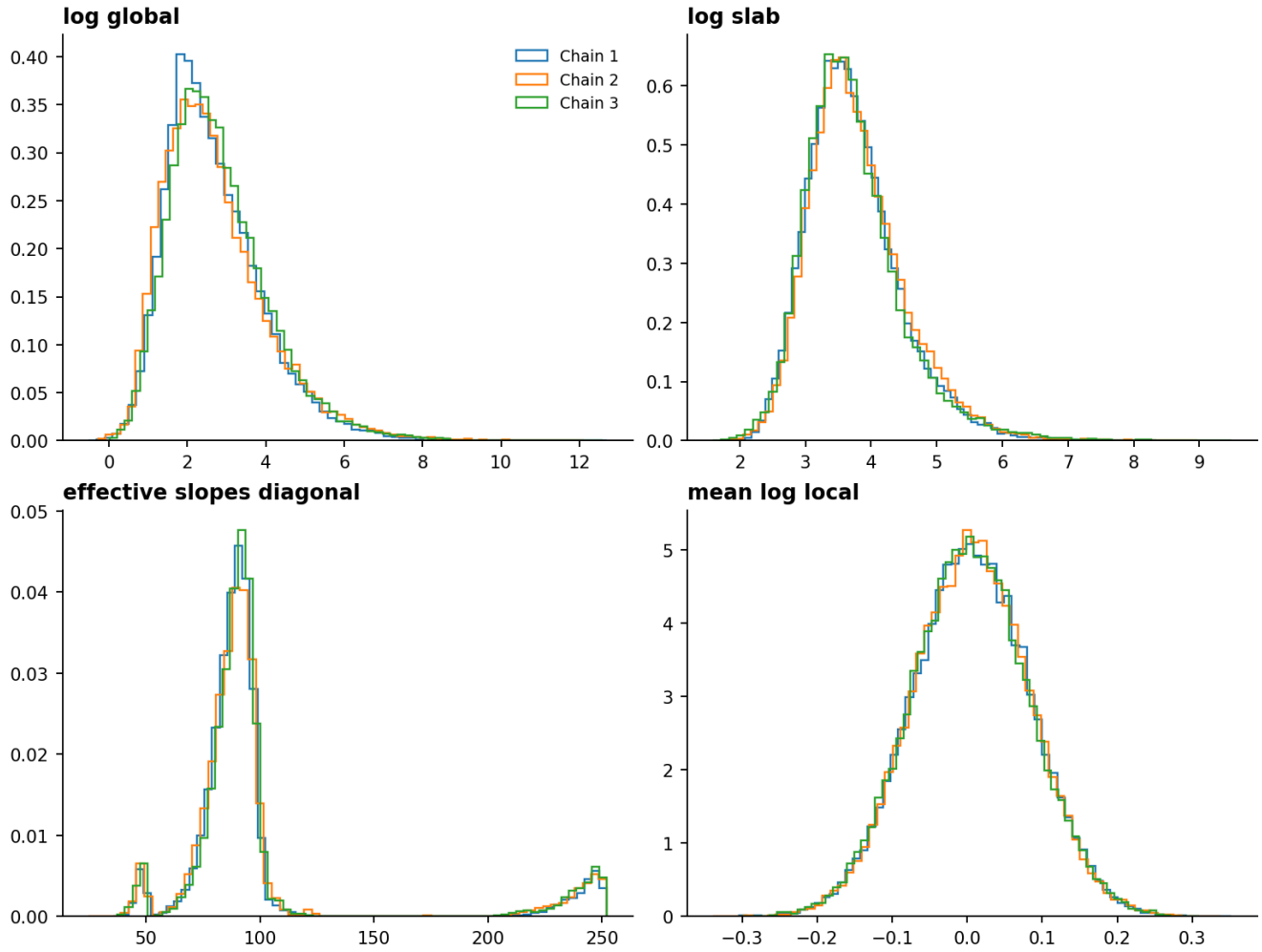
Frozen screens: rank-normalized folded split R-hat <=1.01, summed spectral ESS >=400 for all sampled continuous summaries, and maximum architecture-probability difference between chains <=0.05. Architecture indicator diagnostics are retained separately. Passing screens is evidence of adequate mixing at these budgets, not proof of convergence or exploration of all possible reservoirs.

RHS retained-chain traces



All displayed draws follow the selected attempt's burn-in. Thinning here affects plotting only. Each chain uses a dispersed architecture, observation variance and global-scale initialization. The diagnostic tables use every retained draw.

RHS shrinkage diagnostics



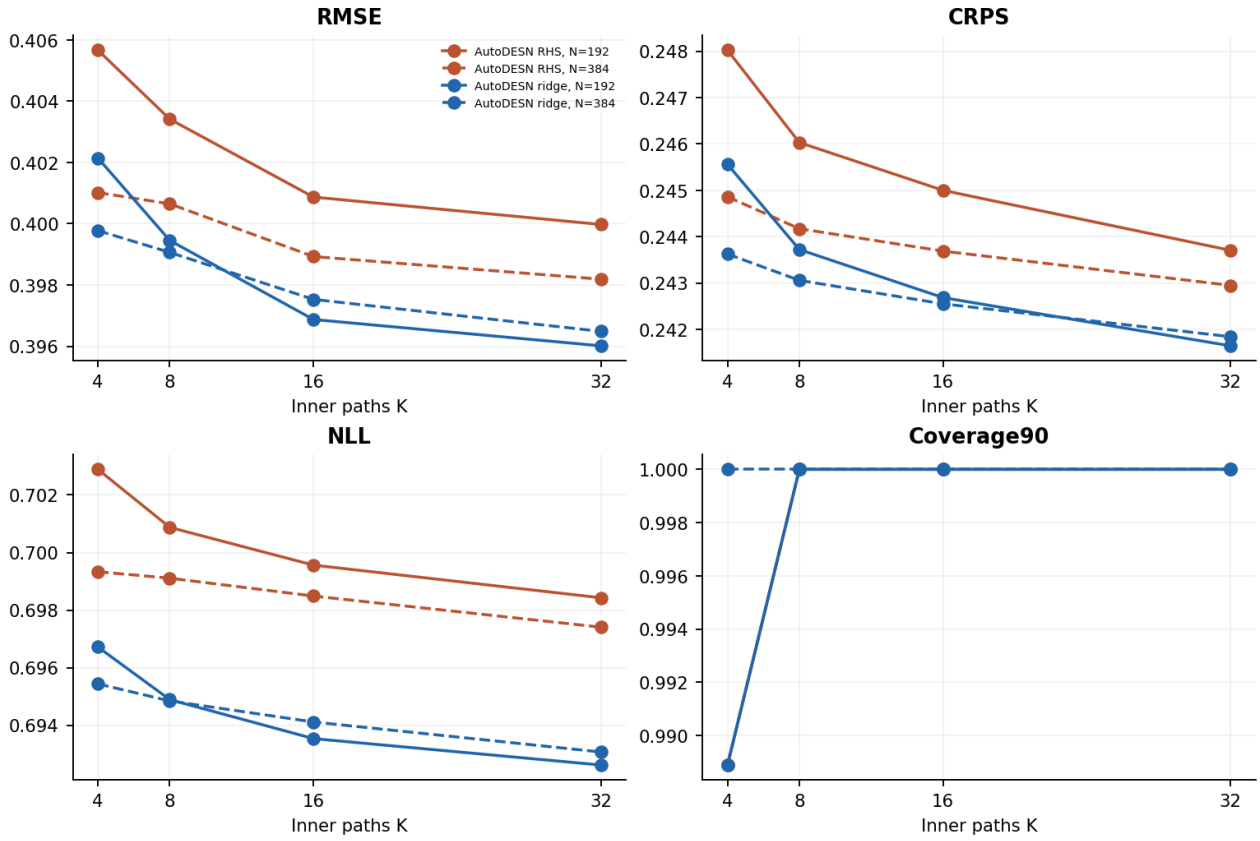
parameter	q05	q50	q95
global_scale	2.8725	12.4526	169.8237
global_tau	2.8725	12.4526	169.8237
slab_variance	15.7023	38.5283	165.1322
effective_slopes_diagonal	65.6250	89.7843	241.3958
mean_log_local	-0.1243	0.0050	0.1352

The effective-slopes statistic uses a diagonal-design approximation; it is not an exact degrees-of-freedom measure with correlated reservoir columns. Mean log local scale averages the common vector, including inactive auxiliary entries. Individual local-scale trajectories were not retained in this large-bank run. Leading-model coefficient summaries are exported separately. Global-scale quantiles and log-scale diagnostics are emphasized: the half-Cauchy global tail need not have a finite posterior mean once the slab caps coefficient variances.

Sensitivity to inference and predictive budgets

method	attempt	qualified	max_rhat	min_ess	leading_probability
ridge	1	True	1.0008	521.9809	0.8061
rhs	1	False	1.0305	252.4428	0.8221
rhs	2	False	1.0108	587.3243	0.8117
rhs	3	True	1.0021	1148.6099	0.8075

Attempts are separate new fits with fixed seeds. Failed attempts are retained for diagnosis and computational accounting; only the qualified attempt supplies forecast draws. A proposal frozen from training occupancy, mixed with 10% original prior mass, preserves full support and uses the exact Hastings correction. It changes computation, not the Bayesian target.



Predictive sensitivity averages three fixed origins and compares N=192/384 joint posterior draws and K=4/8/16/32 inner paths. These are Monte Carlo changes on a fixed fitted posterior, not new training replicates. Main forecasts use N=192 and K=8. The compact posterior contains 384 jointly sampled draws.

Runtime, resources and reproducibility

Model	Phase	Seconds	Scope
Shared DESN bank	Recovered bank-build invocation	128.7040	Both priors
AutoDESN ridge	Inference, all completed attempts	95.2430	3 chains/attempt
AutoDESN ridge	Forecast compute sum	158.9790	471 origins
AutoDESN rhs	Inference, all completed attempts	357.7290	3 chains/attempt
AutoDESN rhs	Training proposal source attempt	44.6950	Additional
AutoDESN rhs	Forecast compute sum	153.8140	471 origins
Native AutoGP	Reused native SMC training	1470.0628	2 x 8 particles
Native AutoGP	Fresh condition/score/export	22.2948	471 origins

Bank construction is shared once by ridge and RHS. Inference timings sum serial chain computations and disclose earlier attempts and proposal preparation. Forecast compute sums per-origin calls; parallel wall time is different. AutoGP training is reused from two pinned native Julia SMC runs, while its rolling conditioning/scoring/export was replayed and numerically matched. These measurements do not establish a hardware-independent speed ranking.

One concurrent attempt exceeded the workspace's 8 GB limit and was killed. Complete chains were preserved, and the final RHS attempt was resumed sequentially. Resource files distinguish that interrupted invocation from completed work. Linux peak RSS measures a process tree maximum and does not sum memory shared by forked workers.

operation	median_seconds_cpp	median_seconds_r	R_over_cpp
complete_bank_fit	0.1330	0.3240	2.4361
Gaussian_model_score	0.0001	0.0003	3.0000
local_scale_update	0.0000	0.0022	50.5227

Paired kernel timings exclude compilation and bank construction. The complete-fit speed ratio is measured on the reference small bank, not inferred from individual-kernel ratios.

Independent validation and practical limits

All 32 independent posterior-reference comparisons passed; maximum quadrature change in architecture probability was 2.86e-06. These references integrate observation-space Gaussian likelihoods and priors independently of the production coefficient-space evidence kernel.

Installed-package validation passed 611 expectations with zero failures or warnings; one unavailable-backend test was skipped because the native DLL was loaded and was covered by the separate source-only run. R study contracts passed 73 expectations and six Python contracts passed. Focused tests cover prior normalization, coefficient moments, local/global/slab updates, detailed balance, architecture odds, unequal readout dimensions, R/C++ parity, seeded reproducibility, numerical stability, API compatibility and future-replacement causality. Separate R and Python study contracts check the frozen protocol and reject incomplete or unqualified outputs. Test logs and the package-check status are included in the evidence directory.

The package tarball installed and its compiled code checked successfully. The final scoped R CMD check (examples, tests and vignettes disabled; implementation tests run separately) reported 1 error, 3 warnings and 2 notes: inherited nonportable validation filenames, non-ASCII legacy R files and legacy help/code mismatches. The newly introduced RHS help cross-reference warning was corrected. This is not a CI-clean package release.

Limits and reproduction

This is a homoscedastic Gaussian model, even when C1 has changing variability. RHS changes coefficient shrinkage; it does not add a dynamic noise model or time-varying coefficients. AutoGP has 2 independent SMC runs of 8 particles each; particle degeneracy and variation between runs limit its own posterior approximation. DESN forecast intervals also have Monte Carlo error.

Run commands, input hashes, source hashes, all origin-horizon scores, diagnostics, budgets, phase timings and checkpoint reconstruction instructions accompany the PR. Repository-wide R CMD check remains distinct from the focused implementation validation; inherited package issues must not be represented as a clean release check.